

Mohamed Khalil Ibrahim Eldesoki

Senior Software Engineer

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M5alil

Fayoum, Egypt

Bio

Senior Software Engineer with over seven years of experience in designing and implementing business-critical software solutions, worked in modernizing legacy systems (20+ years old) through scalable architectures, DevOps CI/CD pipelines, and automated testing frameworks. I have collaborated with global organizations particularly in medical devices to deliver GUI applications and device simulators, develop enterprise-grade software distribution platforms, integrate IoT telemetry with backend systems, and build Tekla Structures plugins for engineering-standard compliance. As a self-driven learner, I translated complex algorithms (e.g., 3D computational geometry) into user-facing applications while staying current with domain-driven design, OAuth/identity management. Holding a Bachelor of Engineering, an AEC Diploma from the ITI, and MBA from EJust.

EXPERIENCE

• Senior dotnet developer

@VAES.ai

August 2025 - present

remote

• Senior software developer

@FlairsTech

July 2023 - July 2025

Maadi, Cairo

Working with FlairsTech has given me the opportunity to work with great organizations around the world like Boston Scientific (BSC). With BSC I am a part of a team that works in the medical devices industry. And I support designing device system software, and other software that relates to the medical system provided by BSC.

- developed medical devices gui and simulators.
- developed appstore in order to achieve proper ways to distribute and update software installed by hospitals and doctors.
- integrate devices to backend and messaging to get data about the device usability and maintenance.
- investigate software update practices for devices and IoT to help using best practices.

• Senior system engineer

@Zamil Steel

June 2018 - July 2023

Maadi, Cairo

played a vital role in a team that was responsible for making and maintaining systems for the company, that provided multiple solutions regarding communication documents, and engineering processes like design, detailing, drawings, and fabrication. I was responsible for:

- maintaining legacy software that search in standard engineering drawings database.
- developing new plugins on Tekla Structures that help detailers save time and keep them in standard details.
- giving training to various companies's offices for using company's systems.
- assisting in developing new software to help with project transformation to Revit.

• Computational geometry developer

@Freelance

2020

remote

working with web technologies in the computational geometry field.

• Structural Engineer

@EGEC

2015 - 2017

Giza

Working as a structural engineer Designing concrete and steel building and prepare Design documents and structural drawings.

EDUCATION

• Egypt-Japan University of Science and Technology (EJUST)

Masters of Business Administration (MBA)

2023 - 2025

• Information Technology Institute (ITI)

Diploma, AEC Informatics technologies

2017 - 2018

• Fayoum University, Faculty of engineering

Bachelor of Engineering

2009 - 2014

TECHNICAL SKILLS AND INTERESTS

Languages: C#, C++, Javascript, Typescript, Python

Tools: Git, TFVC, Visual studio

Libraries and Frameworks: EFCore, Sqlite, MVC, Angular, React, WPF, RxJS, Rx.NET, Three Js

Areas of Interest: DDD, Backend development, OAuth and Identity management,

PROJECTS

- **Cumulocity Integration:** Led team on Investigating the capabilities of Cumulocity and the integration possibility and connectivity.
- **SmartFreeze Simulator:** Worked on a simulator app, that simulates a medical device in action (eg. simulating temperature of catheter while doing operation on patients.) and also worked on updating the App gui.
- **BSC Appstore:** Enterprise appstore that hosts internally developed desktop apps, and Enables admins to assign and approve App download requests to Enterprise users.
- **SmartFreeze Ablation Viewing app:** A desktop app to enable doctors to view and analyze operation gathered data.
- **Eduka AB3D (freelance):** Modernized and migrated a 3D solver to the web, enabling prediction of 3D shapes from 2D projections.
- **Tekla plugins:** Developed new plugins for Tekla Structures to ensure compliance with company standards.
- **Zamil framing tools:** Created backend domain logic and UI for an application that inserts multiple connections and frames.
- **Drawing Creator:** Refactored and solved issues in a legacy tool for creating detailed steel drawings.
- **Design to detailing:** Presented automation ideas to bridge the gap between design and detailing teams.
- **TeklaToRevit:** Developed software to overcome limitations of IFC standards based on company needs.
- **Search Engine for E-Drive:** Maintained MFC C++ legacy application, adding new functionality.
- **ZS-Connection:** Resolved bugs in steel design reports.